

# KakEnBetaal build spec

Oct 2, 2026 · @Richard

## Overview

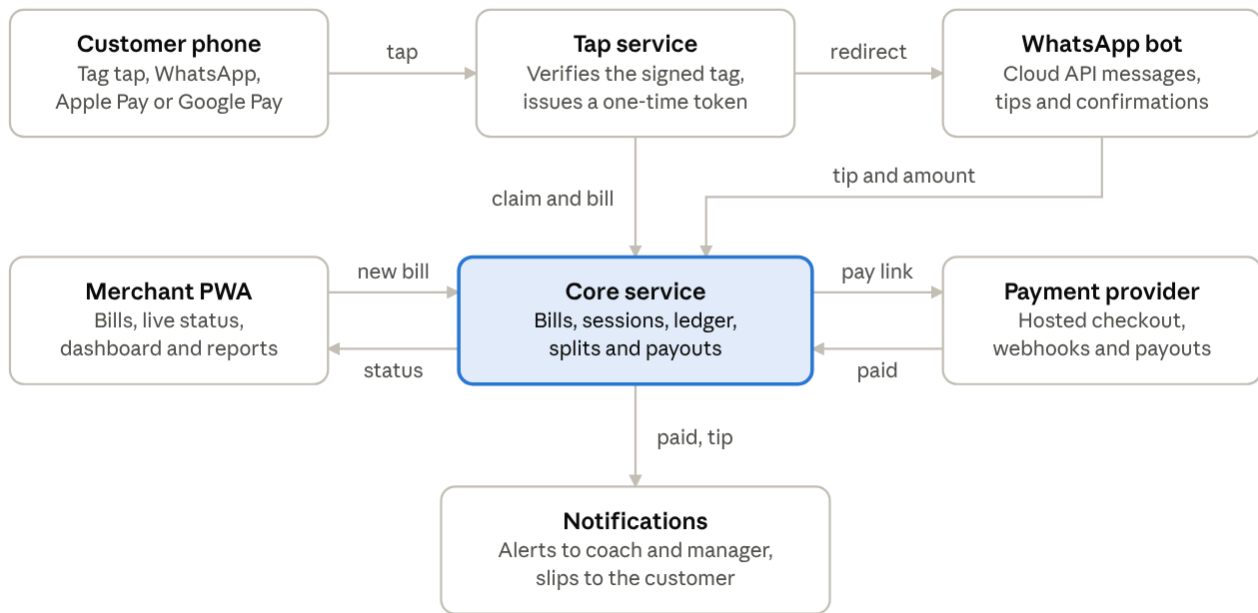
A customer taps an NFC pod, WhatsApp opens with a pre-filled message, and they choose a tip, pay with Apple Pay or Google Pay, and get a branded slip in the same chat. Target: under 30 seconds, no app, no card machine.

Assumptions: a payment provider with hosted checkout and Apple Pay/Google Pay (Peach Payments Checkout supports both), the WhatsApp Cloud API, and one multi-tenant backend serving many merchants.

The tag identifies a coach, till or table and never holds an amount. Merchants create bills in a PWA. The same engine serves six merchant modes (see Merchant modes), and a payment can be split between a shop and the person who served the customer.

1. Tap: the pod's NFC tag opens `wa.me/<number>?text=PAY <tag code>`; the customer taps Send.
2. Bill: the bot looks up the merchant and open bill, then shows items, subtotal and tip choices.
3. Tip: the customer picks No tip, 10%, 15%, 20% or Custom.
4. Pay: the bot sends a Pay Now link to a hosted checkout where the customer approves with Apple Pay or Google Pay.
5. Slip: a webhook confirms payment; the bot sends the receipt, notifies the merchant and queues the tip split.

## One core service connects customer, merchant and payment provider



system architecture · 7 components

The tap service verifies the tag, the bot talks to the customer, and the core service holds bills, sessions and the ledger. The payment provider handles checkout and money movement, and notifications reach merchants and customers.

## Merchant modes

One engine serves every merchant type. Each merchant, or each location, picks a mode in setup. The mode sets who creates the bill, how the customer is matched to it, and the tip and split defaults.

| Mode          | Example merchants                     | Who creates the bill                                         | Customer match                                                     | Tag is bound to | Tip and split default                  |
|---------------|---------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------|-----------------|----------------------------------------|
| Appointment   | Surf coaches, salons, physios, tutors | Coach or booking system, with the customer's WhatsApp number | Number on the bill; a 4-digit bill code if a different number taps | A person        | Shop and coach split; tip to the coach |
| Counter       | Cafes, retail, takeaways              | Till or PWA                                                  | First tap claims the bill                                          | A till          | Optional tip to the pool               |
| Table service | Restaurants, bars                     | Waiter or POS, one bill per table                            | Table tag; each payer claims a share                               | A table         | Tip direct to waiter or pooled         |

| Mode           | Example merchants                      | Who creates the bill                    | Customer match     | Tag is bound to     | Tip and split default              |
|----------------|----------------------------------------|-----------------------------------------|--------------------|---------------------|------------------------------------|
| Quick tip      | Car guards, petrol attendants, buskers | No bill; customer types the amount      | Not needed         | A person            | Preset amounts, 100% to the person |
| Field          | Delivery, on-site trades, markets      | PWA, link sent to the customer's number | Number on the bill | A person or vehicle | Per merchant                       |
| Remote invoice | Anyone billing without a tap           | PWA or dashboard                        | Number on the bill | No tag              | Per merchant                       |

Bill types the engine supports: fixed amount (priced service), open amount (customer or merchant types it), and quick tip (no bill, just an amount to a person).

Per-merchant settings: mode, tip presets, tip split rule, revenue split rule, tag bindings, and bill expiry time. New merchant types are a new combination of these settings, not new code.

## Bill matching and the merchant PWA

The tag only identifies a person, till or table. It never holds an amount. The amount lives on the server as an open bill, and the PWA is how merchants create it.

Matching rules, in order:

1. If the bill has a customer number, send the bill to that number. A tap from the same number confirms it; a tap from another number needs the 4-digit bill code.
2. If the bill has no number, the first customer to tap claims it. The bill locks to that number, and others see "This bill is being paid."
3. The merchant can release a claimed bill from the PWA if the wrong person claimed it.
4. If the merchant edits an amount after a claim, cancel the old session and start a new one so nobody pays a stale amount.
5. A bill link or QR carries the bill ID, so there is no ambiguity when several customers wait.
6. Always confirm merchant, person, service and amount in WhatsApp before Pay Now.
7. Groups: split one bill into shares, or create one bill per person. Each payer claims a share.
8. Someone else may pay a bill. The payer gets the slip, and the merchant sees the payer's number.

Merchant PWA screens:

- Sign in with a PIN or one-time WhatsApp code.

- Pick a service or type an amount, optionally add the customer's WhatsApp number.
- Status: waiting, customer viewing, paid, failed, expired. Shows the merchant's own share on payment.
- Share routes: customer taps the tag, shows a QR code, or opens WhatsApp with the pay link.
- Cached service and price list so it opens on a weak signal; creating a bill needs a connection.

Web NFC only works in Chrome for Android and similar browsers, not iOS. It is used only to write per-sale links to blank tags, which is optional. A coach on iPhone works fully with static tags, QR and WhatsApp send.

## WhatsApp messages

Every message names the merchant. Text in angle brackets is filled in by the backend. Reply buttons are capped at three, so the tip step uses a list message.

| Step            | Who sends             | Message                                                                                              | Buttons or options                  |
|-----------------|-----------------------|------------------------------------------------------------------------------------------------------|-------------------------------------|
| 1. Tap          | Customer (pre-filled) | PAY <one-time token>                                                                                 | Send                                |
| 2. Fixed bill   | Bot                   | Bean and Brew Coffee, Tokai. Beginner lesson with Sipho, R 500.00. Add a tip for Sipho?              | List: No tip, 10%, 15%, 20%, Custom |
| 2a. Quick tip   | Bot                   | Tip Sipho at Bean and Brew. How much?                                                                | Buttons: R 5, R 10, R 20; Other     |
| 2b. Open amount | Bot                   | Enter the amount to pay Bean and Brew Coffee, in rands.                                              | Customer types a number             |
| 2c. Custom tip  | Bot                   | Type your tip in rands, for example 15.                                                              | Customer types a number             |
| 2d. Group share | Bot                   | This bill is R 300.00. Pay your share?                                                               | Equal share, Custom amount          |
| 3. Confirm      | Bot                   | Total R 550.00 (bill R 500.00 + tip R 50.00) to Bean and Brew Coffee for Beginner lesson with Sipho. | Pay now, Change tip                 |

| Step                 | Who sends           | Message                                                                         | Buttons or options                            |
|----------------------|---------------------|---------------------------------------------------------------------------------|-----------------------------------------------|
| 4. Pay               | Bot                 | Opens checkout. Approve with Apple Pay or Google Pay.                           | Link button to hosted checkout                |
| 5. Waiting           | Bot                 | Processing your payment.                                                        | None                                          |
| 6. Success           | Bot                 | Paid R 550.00 to Bean and Brew Coffee. Your slip is below.                      | Slip image or PDF, Get tax invoice            |
| 7. Failed            | Bot                 | That payment didn't go through. You haven't been charged.                       | Try again, Cancel                             |
| 8. Expired           | Bot                 | This payment link has expired. Tap the tag again to restart.                    | None                                          |
| 9. Bill locked       | Bot                 | This bill is being paid from another phone. Ask the merchant to release it.     | None                                          |
| 10. Code needed      | Bot                 | This bill was sent to another number. Enter the 4-digit code from the merchant. | Customer types 4 digits                       |
| 11. No open bill     | Bot                 | There's no open bill for this tag. Ask the merchant to create one.              | None                                          |
| 12. Tag not verified | Bot                 | We couldn't verify this tag. Ask staff to take payment another way.             | None                                          |
| 13. Tax invoice      | Bot                 | Send your company name and VAT number for a tax invoice.                        | Customer replies                              |
| 14. Help             | Customer sends HELP | Bot                                                                             | Pay a bill, Get my slip, Talk to the merchant |

#### Message rules:

- Show the merchant's verified name and the exact total before every Pay Now.
- Never ask for card details in chat. All card entry stays in Apple Pay, Google Pay or the hosted checkout.
- One bill, one link: a link works once and expires after 10 minutes (assumption; tune it).

- Keep every message under 3 lines and use rands with two decimals.

## Tip split

The tip is charged inside the single payment, then separated in the ledger and paid out later. The merchant picks one rule per location in setup.

| Rule      | How it splits                                             | Best for                          |
|-----------|-----------------------------------------------------------|-----------------------------------|
| Direct    | 100% to the staff member tied to the bill or table        | Waiters, hairdressers, car washes |
| Pool      | Equal share, or by hours worked, across everyone on shift | Busy restaurants, cafes           |
| House cut | A set percentage to the owner, the rest by Direct or Pool | Merchants who need to cover costs |

How the money moves:

1. At payment, the ledger records bill amount and tip amount as separate lines on one transaction.
2. Staff are linked to a bill by a staff code entered on the pod or merchant app, or by table assignment. No staff code means the tip goes to the pool.
3. Processing fees on the tip are shared in proportion to the tip, so staff see a net amount. Show the fee rule to staff in plain words.
4. Tips accrue in a per-staff balance. Payout runs daily or when a balance passes a threshold (for example R 100, an assumption), by EFT through the payment provider's payouts product.
5. Staff get a WhatsApp message: "You earned R 11.80 in tips today at Bean and Brew."
6. Refunds reverse the tip proportionally. If the tip is already paid out, the amount is netted off the next payout.

Open question: tax treatment of tips for staff and merchants. Confirm with an accountant before launch.

## Backend steps

1. **Provision tags.** Use NTAG424 DNA tags that sign each tap. Store a per-tag key derived from a master key in a secrets store, and bind the tag to a merchant and a person, till or table. The tag URL points to our own domain, not to WhatsApp directly.

2. **Verify the tap.** Our tap endpoint checks the signature and the tap counter, so cloned or replayed taps are rejected. It creates a one-time claim token valid for about 2 minutes and redirects to `wa.me/<number>?text=PAY <token>`. If WhatsApp isn't installed, it shows a web page with the same pay flow.
3. **Receive the WhatsApp message.** The Cloud API webhook delivers `PAY <token>`. Verify the webhook signature, deduplicate by message ID, and look up the token.
4. **Resolve merchant, mode and bill.** Load the merchant and tag binding. For a fixed bill, apply the matching rules (number match, bill code or first-tap claim) and claim the bill with an atomic database update. For an open bill, ask for the amount. For a quick tip, skip the bill.
5. **Create the session.** Store bill or share, merchant, payer number, a unique session ID and an idempotency key. State `claimed`, expiry 10 minutes (assumption; tune it).
6. **Take the tip or amount.** Handle list replies and typed amounts. Validate: numeric, two decimals, between a minimum and a sane cap. Then show the confirmation message.
7. **Create the checkout.** On Pay now, call the provider API with the total, a reference equal to the session ID and a webhook URL. Return the hosted checkout link in a button.
8. **Confirm payment.** On the provider webhook, verify its signature, match the reference, check amount and currency equal the session total, and make the update idempotent. State moves to `paid`. Never trust the browser redirect alone.
9. **Write the ledger.** Record bill amount, tip, provider fee and the split lines per the merchant's rules, all as integer cents summing to the payment.
10. **Run the split and settlement** per the chosen option (native split, or collect then payout). Failures retry and alert the operator without undoing the payment.
11. **Send slip and notifications.** Render the receipt, send it in WhatsApp, and notify the merchant, coach and manager (see Notifications). Generate the VAT tax invoice on request.
12. **Pay out.** Tip balances and merchant shares pay out on the schedule set for each merchant.
13. **Refunds and disputes.** A merchant triggers a refund in the dashboard. The provider reverses the card payment, and the ledger reverses the split lines. A chargeback notice from the provider does the same and flags the merchant.
14. **Reconcile nightly.** Compare ledger totals with the provider's settlement report and flag any differences.
15. **Monitor.** Track webhook failures, tap-to-slip time, payment success rate and notification delivery, and alert on spikes.

Merchant PWA API:

- `POST /bills` creates a bill (service or amount, optional customer number) and returns its status link.
- `POST /bills/:id/cancel` and `POST /bills/:id/release` cancel a bill or release a wrong claim.
- `GET /bills/:id/events` is a live stream (server-sent events or WebSocket) so the PWA shows paid within seconds.
- Short-lived access tokens after PIN or WhatsApp code sign-in; manager actions need the manager role.

Security checks on every step: per-merchant rate limits on taps, one active claim per tag, the 4-digit bill code limited to 3 attempts, signed webhooks, encrypted secrets, and an audit log of every state change.

## Data model

Keep amounts as integer cents. Every table carries a merchant ID so one backend serves many tenants.

| Table     | Key fields                                                                                                                                             | Purpose                                     |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| merchants | id, name, mode, VAT number, KYC status, provider account refs, default tip and revenue rules, bill expiry                                              | Tenant record and settings                  |
| users     | id, merchant ID, role (owner, manager, staff), WhatsApp number, PIN hash, status                                                                       | Managers, coaches, cashiers, tip recipients |
| devices   | id, user ID, push subscription, last seen, revoked at                                                                                                  | PWA sessions and notifications              |
| tags      | code, merchant ID, bound type (person, till, table), bound ID, key reference, last counter, status                                                     | NFC identity and replay protection          |
| services  | id, merchant ID, name, price, split override                                                                                                           | Priced items such as Beginner lesson        |
| bills     | id, merchant ID, type (fixed, open, quick tip), items, subtotal, tag ID, customer number, claimed by number, bill code, status, expires at, created by | What the customer owes                      |

| Table          | Key fields                                                                                             | Purpose                                      |
|----------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------|
| bill_shares    | id, bill ID, amount, payer number, status                                                              | Group bills                                  |
| sessions       | id, bill or share ID, payer number, state, tip, total, checkout reference, idempotency key, expires at | One payment attempt                          |
| payments       | id, session ID, provider reference, status, method, fee                                                | Provider result                              |
| split_rules    | id, merchant ID, type (direct, pool, house cut, revenue split), shares                                 | Who gets what                                |
| ledger_entries | id, payment ID, type, amount, party ID                                                                 | Bill, tip, fee, split and payout lines       |
| shifts         | id, merchant ID, date, member user IDs, hours                                                          | Tip pools                                    |
| payouts        | id, party ID, amount, status, date                                                                     | Money sent out                               |
| receipts       | id, payment ID, file URL, tax invoice flag                                                             | Slips sent to customers                      |
| notifications  | id, recipient ID, channel, template, payment ID, status, sent at                                       | What each person was told                    |
| webhook_events | id, source, event ID, payload, processed at                                                            | Idempotency for WhatsApp and provider events |
| audit_log      | id, actor, action, target, time                                                                        | Every state change                           |

Bill states: open, claimed, paid, cancelled, expired. Session states: claimed, awaiting\_amount, awaiting\_tip, awaiting\_confirm, awaiting\_payment, paid, failed, expired, cancelled, refunded.

Money rule: amounts are integer cents. Percentage splits are rounded down per party and the remaining cent goes to the merchant, so the lines always sum to the payment.

## Edge cases

| Case                                     | What happens                                                                                                  |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Customer taps twice                      | Reuse the open session for that customer and merchant; don't create a second bill                             |
| Two customers tap at once                | Atomic claim; the second customer gets the bill locked message                                                |
| Wrong person claimed the bill            | Merchant releases it from the PWA; the old session is cancelled                                               |
| Merchant edits the amount after a claim  | Cancel the old session, start a new one, tell the customer                                                    |
| Payment succeeds but the webhook is late | Poll the provider after 60 seconds; send the slip once status is confirmed                                    |
| Webhook arrives twice                    | Idempotent update keyed on provider reference; no second slip                                                 |
| Customer pays twice                      | Unique provider reference per session; the second payment is refunded automatically and both parties are told |
| Amount mismatch                          | Don't mark paid; alert the merchant and the operator                                                          |
| Customer abandons at checkout            | Session expires; send nothing further unless they tap again                                                   |
| Cloned or replayed tag                   | Signature or counter check fails; show the tag not verified message                                           |
| Tag unbound, lost or disabled            | Show the no open bill message; the manager can disable the tag and issue a new one                            |
| Sticker placed over a real tag           | Merchant name in the confirmation lets the customer spot it; tags can be rotated                              |
| Weak signal at the pod                   | The tap itself needs no data; the WhatsApp steps resume when signal returns                                   |
| Merchant offline when the customer pays  | Payment still completes; the merchant is notified when back online                                            |
| Customer types a huge or odd tip         | Validate against the cap and ask again                                                                        |

| Case                               | What happens                                                                                            |
|------------------------------------|---------------------------------------------------------------------------------------------------------|
| Split transfer fails after payment | The payment stays paid; retry the transfer; show the recipient's balance as pending; alert the operator |
| Refund requested                   | Merchant triggers it in the dashboard; the provider refunds; split lines reverse                        |
| Refund after payout                | Net off the next payout, or ask the merchant to cover it                                                |
| Chargeback                         | Provider notifies; reverse the split lines; flag the merchant                                           |
| Staff member leaves                | Disable the user, device and tag; pay out any balance to the bank on file                               |
| Phone lost or stolen               | Manager revokes the device; PIN locks after 5 wrong tries                                               |
| Provider outage                    | Show a payments-down message; bills stay open; no new links                                             |
| WhatsApp outage                    | The PWA offers the pay link or QR code with a web checkout page                                         |
| WhatsApp 24-hour window closed     | Tapping reopens it; for later messages use an approved template                                         |
| Notification not delivered         | Fall back along the channel chain; mark the failure in the dashboard                                    |

## Compliance and open questions

This is a planning checklist, not legal advice. Get a payments lawyer to review it before launch.

- ☐ Payment provider agreement: does it allow you to onboard other merchants (sub-merchants or a payment facilitator model), and who is the merchant of record?
- ☐ Holding funds: if you hold merchant or tip money, you may need a bank sponsor or a payment licence. Split settlement straight to merchants avoids this.
- ☐ Card data: customers enter card details only in Apple Pay, Google Pay or the provider's hosted page, keeping you out of card-data scope.
- ☐ POPIA: consent for storing WhatsApp numbers and receipts, a retention period, and a way to delete data on request.

- ☐ **Meta:** a verified business account, approved message templates, and the tap opening the customer's service window.
- ☐ **Tax:** VAT tax invoice format for business customers, and the treatment of tips for staff and merchants.
- ☐ **Provider fee:** confirm the exact card fee, and whether Pay by Bank or PayShap can be offered as a cheaper option.
- ☐ **NFC hardware:** tag type, price per pod, and whether the pod updates the URL per sale or only identifies the merchant.

## Notifications

Merchants must know instantly that a bill was paid and tipped. The tag gives no feedback, so the phone has to say it. Notify only when the provider's webhook confirms payment, never when the customer taps Pay now.

| Who                 | When                             | Message                                                                                 |
|---------------------|----------------------------------|-----------------------------------------------------------------------------------------|
| Coach or cashier    | Payment confirmed                | Paid R 500.00 (Beginner lesson) + R 50.00 tip. Your share R 400.00. Customer ending 482 |
| Coach or cashier    | Failed or expired                | Payment for Beginner lesson didn't go through. Tap Retry to resend                      |
| Shop manager        | Each payment, or a daily summary | Total, shop share, coach name, tip amount                                               |
| Staff on a tip pool | At payment and at payout         | You earned R 11.80 in tips. Paid out R 142.50 today                                     |
| Everyone involved   | Refund                           | R 500.00 refunded; your share reverses by R 400.00                                      |

Channels, in order of priority:

1. PWA live status: the bill screen flips to Paid with a sound and a vibration while the app is open.
2. Push notification: works on Android Chrome; on iPhone the PWA must be installed to the home screen first, so don't rely on it alone.
3. WhatsApp message: the backup that reaches everyone. Business-initiated messages need approved templates and may be charged; check Meta's current rates.
4. SMS: last resort for merchants without data.
5. Smart pod (later): an LED and a beep so the customer sees Paid at the tag too.

## Rules:

- Show tip and bill amounts separately; the tip is the figure staff care about most.
- Mask the customer's number (last 3 digits) or show a first name only; POPIA applies.
- One notification per payment, keyed on the provider reference.
- If a push isn't delivered, send WhatsApp; if WhatsApp fails, mark it in the dashboard.
- Managers choose who gets which alerts and set quiet hours; coaches can mute sounds.
- Log every notification with its channel and delivery status.

## Operations

### Merchant onboarding:

1. The shop signs up and completes business verification (KYC) with the payment provider; collect bank details and VAT status.
2. The manager adds services and prices, tip presets, and tip and revenue split rules.
3. The manager adds staff by WhatsApp number; each gets a tag and installs the PWA.
4. Run a test payment for every tag and every notification channel.
5. Go live; the operator watches the first week of payments.

### Support and money operations:

- Customers can message HELP for a menu: pay a bill, get a slip, contact the merchant.
- Merchants get a support WhatsApp number and a documented path for failed payments, wrong amounts and refunds.
- The operator console shows merchants, tag inventory, failed webhooks, unreconciled payments and pending payouts.
- Daily: reconcile the ledger with the provider's settlement report and review flagged payments. Weekly: review payouts, refunds and chargebacks.

## Non-functional requirements and testing

Targets below are assumptions to confirm in the pilot.

| Requirement                                | Target          |
|--------------------------------------------|-----------------|
| Tap to bill message                        | Under 3 seconds |
| Payment confirmed to merchant notification | Under 5 seconds |

| Requirement                   | Target                                                                               |
|-------------------------------|--------------------------------------------------------------------------------------|
| Availability during the pilot | 99.5% per month                                                                      |
| Pilot scale, with headroom    | 10 merchants and 1,000 payments per day; design for 100 merchants without a redesign |
| Money handling                | Integer cents, audit log of every state change, daily backups                        |
| Customer phones               | iPhone XS or later (reads tags in the background); Android with NFC switched on      |
| Merchant phones               | Any phone for the PWA; Chrome on Android for optional tag writing                    |
| Privacy                       | POPIA: consent, retention period, deletion on request                                |

#### Test plan:

- Unit tests for money maths, especially rounding and splits that must sum to the payment.
- End-to-end sandbox tests for every merchant mode.
- A device matrix: several iPhone and Android models, wet and dry tags, weak signal.
- Failure injection: duplicate and late webhooks, provider down, WhatsApp down, notification failures.
- Security tests: cloned and replayed tags, bill code guessing, webhook forgery, expired links.
- A load test at several times the pilot volume.
- A dry run with real small payments before the pilot opens.

## Revenue split and tag hardware

Revenue split. A payment can be split between the shop and the person who served the customer. Example: a R500 lesson plus a R50 tip is one customer payment of R550. The shop gets R150 (30% of the lesson), and the coach gets R350 plus the R50 tip.

#### Three ways to split:

1. Gateway-native split: PayFast Split Payments instantly splits part of a payment to a second PayFast account, through custom integration. An older forum thread says recipients must be PayFast accounts; check this.

2. Collect, then pay out: pay everyone from one account through a payouts API (Peach Payments has a payouts product). This means holding other people's money, so get legal advice first.
3. Marketplace or sub-merchant accounts: ask Peach whether split settlement to sub-merchants with provider-run KYC is available in South Africa. Not confirmed.

Recommendation: prototype with option 1 if the shop and coaches can open PayFast accounts, ask Peach about option 3 in parallel, and use option 2 only after legal review. The shop is the merchant of record, so refunds go through the shop; decide who absorbs the processing fee.

Tag hardware. Prices before shipping, 15% VAT and duty; plan for 30% to 50% on top (an estimate), at about R16.5 to R17 per US dollar (July 2026 rate).

| Option                     | Chip        | Price at volume                 | Approx. rands each |
|----------------------------|-------------|---------------------------------|--------------------|
| Printed silicone wristband | NTAG213     | \$1.69 at 500, \$0.95 at 10,000 | R28 down to R16    |
| Signed sticker tag         | NTAG424 DNA | \$1.03 at 100, \$0.96 at 1,000  | R17 to R16         |
| Signed PVC card            | NTAG424 DNA | \$2.50 at 100+                  | R41                |

NTAG213 is fine for a pilot but cannot sign each tap, so a copied tag works. Use NTAG424 DNA in production so the server can reject cloned or replayed taps. A signed sticker on a plain silicone band or keyfob is a practical signed wristband. No published price found for a silicone wristband with a signed chip; request quotes.

## Unpaid bills and reminders

A customer may tap and leave, or a payment may fail. The bill then stays open, and the bot reminds the customer a limited number of times before handing the bill to the merchant.

| When                                     | Message                                                     | Notes                |
|------------------------------------------|-------------------------------------------------------------|----------------------|
| About 10 minutes after the customer left | You haven't finished paying<br><merchant> R 550.00. Pay now | Fresh pay link       |
| Next morning                             | A second short reminder                                     | Only if still unpaid |

| When         | Message                                   | Notes                                                               |
|--------------|-------------------------------------------|---------------------------------------------------------------------|
| After 3 days | A final reminder that says it is the last | Then the bill moves to the merchant's Unpaid tab as needs follow-up |

Reminders stop immediately when the customer pays by any route, replies STOP, the merchant cancels or writes off the bill, someone else pays it, or the merchant turned reminders off for that service.

Fair-use rules: the first message tells the customer they will be reminded and how to stop; every reminder names the merchant and amount and offers an opt-out; send only between 08:00 and 20:00; at most one reminder a day and three in total per bill; log every reminder; reminders outside the free 24-hour window need approved templates and may cost money.

Bill statuses added: `abandoned`, `needs_follow_up`, `written_off`, `paid_other` (cash or EFT, with a logged reason). New data: a `reminders` table (bill, session, sequence number, sent at, template, status) and an `opt_outs` table (hashed number, merchant, scope).

Merchant controls: number of reminders (0 to 3), timing, quiet hours, an Unpaid tab with Send reminder, Resend link, Mark paid another way and Write off. Prevention: take a deposit at booking, use pay-first at counters, or ask your provider about card holds.

## Customer details and privacy

The customer's chat goes to the product's business number, not to the merchant, so merchants see details second-hand.

| Where             | What the merchant sees                                                                |
|-------------------|---------------------------------------------------------------------------------------|
| WhatsApp alert    | Paid R 550.00 (Beginner lesson), tip R 50.00, your share R 400.00, Siphos, ending 482 |
| PWA bill screen   | Status, amount, tip, first or profile name, masked number                             |
| Unpaid tab        | The same, plus reminders sent                                                         |
| Dashboard reports | Totals by day, coach and tip, with no personal details                                |

The system receives the customer's WhatsApp number and profile name, the time, merchant, bill and how far they got. It never receives card details or a surname.

Number visibility: appointment merchants enter the number from the booking; walk-ups show a first name and masked number; for unpaid bills the first message offers a Yes or No button to share the number with the merchant, and the full number is revealed only on Yes. Confirm the wording and retention period with a lawyer (POPIA).

## Payment methods and wallets

| Phase | What                                                                   | Notes                                                                                                                                                                       |
|-------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Apple Pay, Google Pay, Pay by Bank or Capitec Pay through the provider | Account-to-account methods may cost less than cards; Samsung Pay needs its own sign-up                                                                                      |
| 2     | Saved payment method (provider token), one tap for repeat customers    | Confirm tokenization with the provider                                                                                                                                      |
| 3     | Merchant store credit and lesson packs                                 | Merchant owes the credit; check limits with a lawyer                                                                                                                        |
| 4     | A stored-balance eWallet                                               | Only through a bank partner or after authorisation; only banks may issue e-money in South Africa today and the SARB framework for non-banks was still in draft when checked |

Staff tips: prefer split settlement straight to the staff member over holding a balance.

## Recommended pilot plan

- No eWallet yet; offer the provider's wallets and Pay by Bank.
- Pilot with a surf shop (2 to 3 coaches, appointment mode, revenue split, tips, unpaid bills) and a cafe (counter mode).
- Choose the provider by written confirmation of split support and fees: PayFast native split or Peach sub-merchant split.
- Do not hold anyone's money; settle directly. See a payments lawyer once, on the funds flow.
- About 50 signed NTAG424 DNA stickers on plain silicone bands, with a QR fallback.
- Defer a smart pod, table service, group bills and venue-wide tip pools.

## Build order

| Milestone                  | Scope                                                           | Done when                                                                                                                                     |
|----------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Prototype               | One merchant, one tag, fixed bill, provider sandbox             | A test customer on iPhone and Android goes from tap to slip in under 30 seconds in the sandbox                                                |
| 2. Core modes              | Appointment and counter modes, bill matching, claim and release | Two phones tapping one bill: exactly one succeeds; a wrong claim can be released; no double charge                                            |
| 3. Tips                    | Tip step, quick tip mode, custom tip                            | Tip lines are recorded separately; every preset and custom amount is tested                                                                   |
| 4. PWA and notifications   | Bill creation, live status, push and WhatsApp alerts            | The coach sees Paid within 5 seconds of confirmation on Android and iPhone; no duplicate alerts                                               |
| 5. Money                   | Split rules, ledger, payouts, refunds                           | For 100 random test payments the split lines sum to the payment to the cent; payouts reconcile; a refund reverses every line                  |
| 6. Merchant dashboard      | Services, staff, tags, reports, refunds, tax invoice            | A manager onboards a coach and issues a tag without your help                                                                                 |
| 7. Security and compliance | Signed tags, rate limits, POPIA, legal review of the funds flow | Cloned and replayed tags are rejected; every compliance checklist item is closed                                                              |
| 8. Pilot                   | 2 to 3 merchants for 2 to 4 weeks                               | Targets met: median tap to slip under 30 seconds, payment success above 90%, zero unreconciled payments (all targets are assumptions to tune) |

Measure in the pilot: seconds from tap to slip, drop-off at the tip step, drop-off at checkout, the share of customers who add a tip, and how often merchants need to release or cancel a bill.

## Open decisions

- ☐ First pilot merchants: which mode (appointment, counter, table, quick tip)?
- ☐ Who creates bills at each pilot merchant?
- ☐ Split option: PayFast native, Peach sub-merchants, or collect and pay out?
- ☐ Tag type for the pilot (NTAG213 or signed sticker) and quantity.
- ☐ How the platform earns: a per-transaction markup, a monthly merchant fee, tag sales, or a mix?
- ☐ Who owns the verified WhatsApp Business account and phone number, and what is the product's name?
- ☐ Which payment provider handles split settlement, and has its sales team confirmed it in writing?